CLAIMS

WHAT IS CLAIMED IS:

1. A control method for a cleaning robot, comprising:

receiving a weekly cleaning constraining condition input by a user, wherein the weekly cleaning constraining condition represents a condition to be satisfied when the cleaning robot cleans, in a pre-set period, one or more sub-zones of a plurality of sub-zones corresponding to a work zone included in a map of the work zone, wherein the pre-set period is an integer multiple of a week;

determining weekly plan options that satisfy the weekly cleaning constraining condition;

providing the weekly plan options to the user;

obtaining a to-be-executed weekly plan option selected by the user from the weekly plan options; and

controlling the cleaning robot to execute a cleaning task in the pre-set period according to the to-be-executed weekly plan option.

2. The control method of claim 1, wherein the weekly cleaning constraining condition includes a first condition and/or a second condition, wherein the first condition relates to a number of cleanings to be performed in the pre-set period, and the second condition relates to an area to be cleaned in the pre-set period.

3. The control method of claim 1, wherein the weekly cleaning constraining condition also includes a third condition, wherein the third condition relates to a cleaning time in which cleaning is to be performed in the pre-set period.

4. The control method of claim 1, wherein the weekly cleaning constraining condition also includes a fourth condition, and wherein the fourth condition relates to a cleaning order in which cleanings are to be performed in the pre-set period.

5. The control method of claim 1 wherein the weekly cleaning constraining condition also

includes a fifth condition, and wherein the fifth condition relates to a moving mode in the pre-set period.

6. The control method of claim 2,

wherein the plurality of sub-zones are divided into a plurality of categories, and

wherein the first condition is configured to specify a total number of cleanings to be performed in the pre-set period for a corresponding category of sub-zones and/or to specify a number of cleanings to be performed on each day of the pre-set period for a corresponding category of sub-zones.

7. The control method of claim 2,

wherein the plurality of sub-zones are divided into a plurality of categories,

wherein the second condition is configured to specify a first area that is a sum of areas of sub-zones of all categories in which cleanings are to be performed in the pre-set period and/or a second area that is a sum of areas of sub-zones of all categories in which cleanings are to be performed on a single day,

wherein the first area is equal to a sum of products, wherein each product is a product of an area of each category of sub-zones and a number of cleanings to be performed in the pre-set period corresponding to each category of sub-zones, and

wherein the second area is equal to a sum of products, wherein each product is a product of an area of each category of sub-zones in which cleaning is to be performed on a single day and a number of cleanings to be performed on a single day for each corresponding category of sub-zones.

8. The control method of claim 3, wherein the third condition is configured to specify at least one of the following: a number of interval days between cleanings for a corresponding category of sub-zones, a cleaning time of a pre-set day of the pre-set period being the shortest, or a do-not-disturb time segment in the pre-set period.

9. The control method of claim 4, wherein the fourth condition is configured to specify an order in which a plurality of sub-zones of a same category are to be cleaned and/or an order in which sub-zones of a plurality of categories are to be cleaned on a same day.

10. The control method of claim 5, wherein the moving mode includes at least one of a cornrow covering mode, an edge following mode, a predicament escaping mode, a navigation mode, or a return-to-recharge mode,

wherein the return-to-recharge mode includes a first return-to-recharge mode, a second return-to-recharge mode, and a third return-to-recharge mode,

wherein in the first return-to-recharge mode, the cleaning robot performs return-to-recharge after having moved past a pre-set time segment or after having moved for a pre-set amount of moving time,

wherein in the second return-to-recharge mode, the cleaning robot performs the return-to-recharge when a remaining electrical power is less than a pre-set electrical power threshold,

wherein in the third return-to-recharge mode, the cleaning robot performs return-to-recharge after having moved past a pre-set time segment or after having moved for a pre-set amount of moving time, or after having cleaned a pre-set area, and after completing recharging, the cleaning robot continues to move until the remaining electrical power is less than the pre-set electrical power threshold, and again performs return-to-recharge and does not execute a cleaning task again after recharging is completed.

11. The control method of claim 1, further comprising:

on a current day of the pre-set period in which a cleaning task is to be executed, determining whether the cleaning robot has already executed the cleaning before a pre-set time on the current day,

wherein when the determination is that the cleaning task has not been executed before the pre-set time on the current day, reminding the user to control the cleaning robot to execute the cleaning task, or bypassing the user and directly controlling the cleaning robot to execute the cleaning task.

12. The control method of claim 1, wherein the weekly plan options change as the weekly cleaning constraining condition changes, and the control method further comprises:

in response to a change in the to-be-executed weekly plan option, controlling the cleaning robot to execute new cleaning task according to to-be-executed weekly plan option in the pre-set period.

13. The control method of claim 1, further comprising:

when there is a conflict between at least two conditions included in the weekly cleaning constraining condition, performing an error processing corresponding to the conflict.

14. The control method of claim 1, further comprising:

based on pre-set or user-set priorities of conditions included in the weekly cleaning constraining condition, determining the weekly plan options based on an order from high priority to low priority.

15. A control apparatus for a cleaning robot, comprising:

a receiving unit configured to receive a weekly cleaning constraining condition input by a user, wherein the weekly cleaning constraining condition represents a condition to be satisfied when the cleaning robot cleans, in a pre-set period, one or more sub-zones of a plurality of sub-zones corresponding to a work zone included in a map of the work zone, wherein the pre-set period is an integer multiple of a week,

a determining unit configured to determine weekly plan options that satisfy the weekly cleaning constraining condition;

a rendering unit configured to provide the weekly plan options to the user;

an obtaining unit configured to obtain a to-be-executed weekly plan option selected by the user from the weekly plan options; and

a control unit configured to control the cleaning robot to execute a cleaning task in the pre-set period according to the to-be-executed weekly plan option.

16. The control apparatus of claim 15, wherein the weekly cleaning constraining condition comprises one or more of the following conditions:

a first condition configured to specify a number of cleanings to be performed in the pre-set period;

a second condition configured to specify an area to be cleaned in the pre-set period;

a third condition configured to specify a cleaning time in which cleaning is to be performed in the pre-set period;

a fourth condition configured to specify a cleaning order in which cleanings are to be performed in the pre-set period; and

a fifth condition configured to specify a moving mode in the pre-set period.

17. The control apparatus of claim 15,

wherein the plurality of sub-zones are divided into a plurality of categories,

wherein the first condition is configured to specify a total number of cleanings to be performed in the pre-set period for a corresponding category of sub-zones and/or to specify a number of cleanings to be performed on each day of the pre-set period for a corresponding category of sub-zones,

wherein the second condition is configured to specify a first area that is a sum of areas of sub-zones of all categories in which cleanings are to be performed in the pre-set period and/or a second area that is a sum of areas of sub-zones of all categories in which cleanings are to be performed on a single day,

wherein the first area is equal to a sum of products, wherein each product is a product of an area of each category of sub-zones and a number of cleanings to be performed in the pre-set period corresponding to each category of sub-zones,

wherein the second area is equal to a sum of products, wherein each product is a product of an area of each category of sub-zones in which cleaning is to be performed on a single day and a number of cleanings to be performed on a single day for each corresponding category of sub-zones,

wherein the third condition is configured to specify at least one of the following: a number of interval days between cleanings for a corresponding category of sub-zones, a cleaning time of a pre-set day of the pre-set period being the shortest, or a do-not-disturb time segment in the pre-set period, and

wherein the fourth condition is configured to specify an order in which a plurality of sub-zones of a same category are to be cleaned and/or an order in which sub-zones of a plurality of categories are to be cleaned on a same day.

18. A non-transitory computer-readable storage medium storing instructions, which when executed by a processor, cause the processor to perform a control method for a cleaning robot, the control method comprising:

receiving a weekly cleaning constraining condition input by a user, wherein the weekly cleaning constraining condition represents a condition to be satisfied when the cleaning robot cleans, in a pre-set period, one or more sub-zones of a plurality of sub-zones corresponding to a work zone included in a map of the work zone, wherein the pre-set period is an integer multiple of a week;

determining weekly plan options that satisfy the weekly cleaning constraining condition;

providing the weekly plan options to the user;

obtaining a to-be-executed weekly plan option selected by the user from the weekly plan options; and

controlling the cleaning robot to execute a cleaning task in the pre-set period according to the to-be-executed weekly plan option.

19. The non-transitory computer-readable storage medium of claim 18, wherein the weekly cleaning constraining condition comprises one or more of the following conditions:

a first condition configured to specify a number of cleanings to be performed in the pre-set period;

a second condition configured to specify an area to be cleaned in the pre-set period;

a third condition configured to specify a cleaning time in which cleaning is to be performed in the pre-set period;

a fourth condition configured to specify a cleaning order in which cleanings are to be performed in the pre-set period; and

a fifth condition configured to specify a moving mode in the pre-set period.

20. The non-transitory computer-readable storage medium of claim 18,

wherein the plurality of sub-zones are divided into a plurality of categories,

wherein the first condition is configured to specify a total number of cleanings to be performed in the pre-set period for a corresponding category of sub-zones and/or to specify a number of cleanings to be performed on each day of the pre-set period for a corresponding category of sub-zones,

wherein the second condition is configured to specify a first area that is a sum of areas of sub-zones of all categories in which cleanings are to be performed in the pre-set period and/or a second area that is a sum of areas of sub-zones of all categories in which cleanings are to be performed on a single day,

wherein the first area is equal to a sum of products, wherein each product is a product of an area of each category of sub-zones and a number of cleanings to be performed in the pre-set period corresponding to each category of sub-zones,

wherein the second area is equal to a sum of products, wherein each product is a product of an area of each category of sub-zones in which cleaning is to be performed on a single day and a number of cleanings to be performed on a single day for each corresponding category of sub-zones,

wherein the third condition is configured to specify at least one of the following: a number of interval days between cleanings for a corresponding category of sub-zones, a cleaning time of a pre-set day of the pre-set period being the shortest, or a do-not-disturb time segment in the pre-set period, and

wherein the fourth condition is configured to specify an order in which a plurality of sub-zones of a same category are to be cleaned and/or an order in which sub-zones of a plurality of categories are to be cleaned on a same day.